Longsnout seahorse

A tail that holds on: the biomechanics behind the Planted Tank's seahorses

Hippocampus reidi · World Foundry aquarium series

verified Captive broodstock averages 15.6–16.6 cm tall.
Largest recorded adult: 17.5 cm. [1, 2]

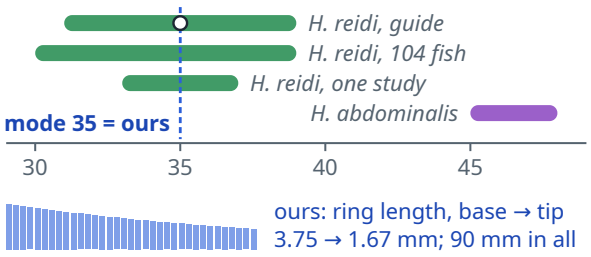
verified Colour: brown spots with tiny white dots, ring stripes, or uniform. [2, 3]

verified Western Atlantic; on seagrass, mangrove roots, macroalgae and gorgonians. [2]

01 · Thirty-five rings



Tail rings: published ranges



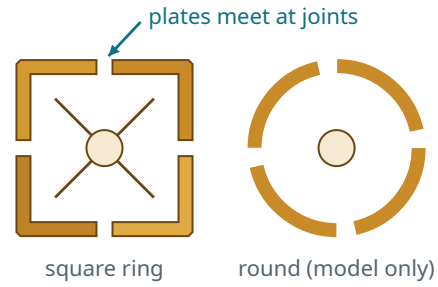
verified 11 trunk rings, and most often 35 tail rings (range 31–39). [2]

verified Other counts: 30–39 tail rings in 104 Brazilian specimens; 33–37 in another study. [3, 4]

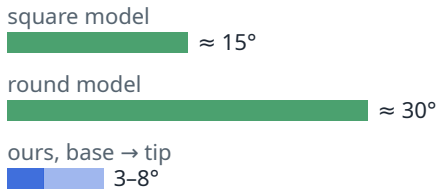
verified Each tail ring is four L-shaped bony plates around one vertebra. [5]

ours Our tail: 35 rings, 90 mm. Rings shrink from 3.75 to 1.67 mm long and 9.0 to 1.4 mm wide.

02 · Square bends without crushing



Twist between neighbouring rings



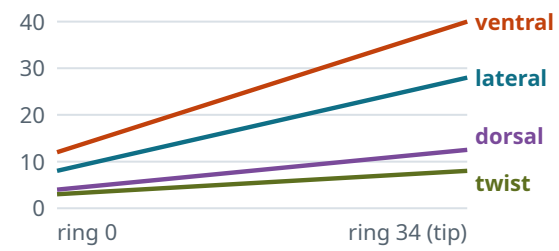
verified Joints: ball-and-socket between vertebrae, peg-and-socket between plates, gliding where plates overlap. [5]

verified In 3D-printed models, the square tail resists crushing better than a round one, and it twists about 15° per segment (round: about 30°). [5]

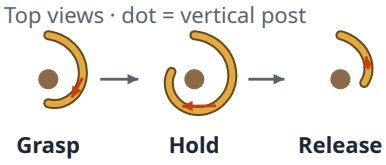
ours Our twist limit rises from 3° at the base to 8° at the tip.

03 · Grip, hold, release

Our limit per joint (°)



Runtime grip (ours)



Tip closes first; release unwinds.

Release → swim → probe → grasp

verified *H. reidi*: whole-tail grip on a 1 cm perch; more motion toward the tip, with sideways bend as it curls. [4]

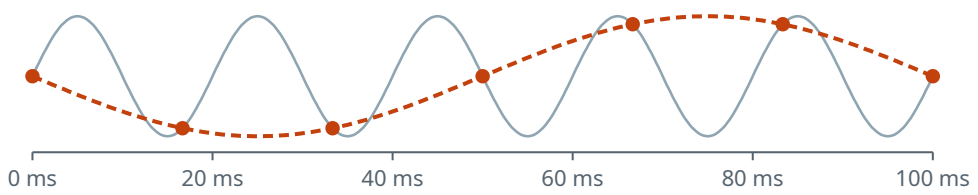
other species *H. abdominalis* chose vertical, round, smooth holdfasts. [4]

ours Our wrap closes from the tip, tracks the post during sway, and unwinds before swimming.

04 · A fin too fast to draw

A 50 Hz fin beat, seen by a 60 Hz screen

Red dots: one per 16.7 ms frame. Joined up, they look like a 10 Hz wave.



Six pairs, close idle: 14.6 FPS, one frame every 68.5 ms.

verified *H. reidi* fins: 20–60 Hz, slowest when hovering; most often about 50 Hz. [6, 7]

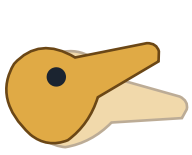
other species *H. erectus* (as *H. hudsonius*) dorsal fin: 30–42 Hz; dwarf seahorse: 42–54 Hz. [8]

other species *H. kuda* holds at most 3–4 cm/s against a current. [9]

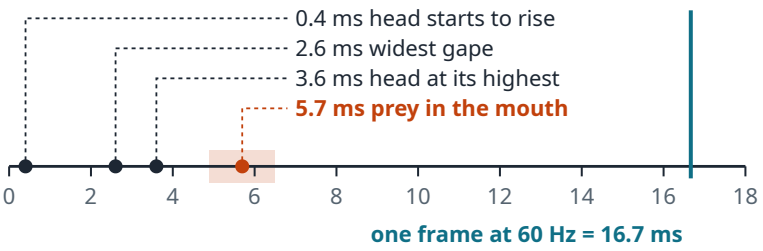
ours A 60 Hz screen shows at most 30 Hz, so we draw a slowed, blurred wave. Swim speed is ours.

05 · A six-millisecond strike

Adult strike, timed from the first hyoid movement (ms)



peak head rotation
26.7 ± 8° [10]



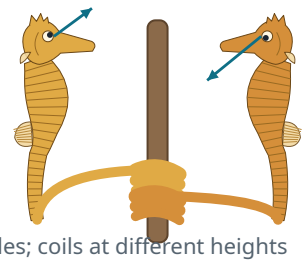
verified Adult *H. reidi* pivot feeding: the head starts to rise 0.4 ms after the hyoid moves; the prey crosses the mouth at 5.7 ± 0.8 ms. [10]

verified Newborns turn the head more than 40° in under 2.5 ms. [11]

ours The strike ends inside one 16.7 ms frame at 60 Hz, so the game cuts from aim to strike.

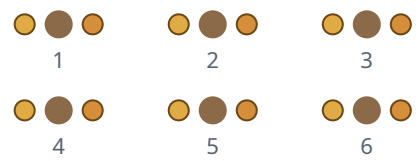
06 · Six pairs share six holdfasts

Greeting (ours)



Opposite sides; coils at different heights

12 residents · 6 holdfasts (ours)



Greeting rotates through visible pairs.
Check occupied spots before moving.

verified *H. reidi* is found in pairs in the wild. [2]

other species Seahorses and pipefishes move each eye on its own to find, track and strike single prey. [12]

ours 12 residents, 6 posts. Greetings rotate through visible pairs and check other residents' spot reservations. Partners share a post at different coil heights; this and independent eye targets are ours.

07 · The model we built

1,101 triangles / seahorse

2 actors: body + fins

693 authored clip frames

35 rings · flex + sideways bend + twist · 7 control knots



21 knot angles + 4 wrap values → **105 joint angles + clip blending**

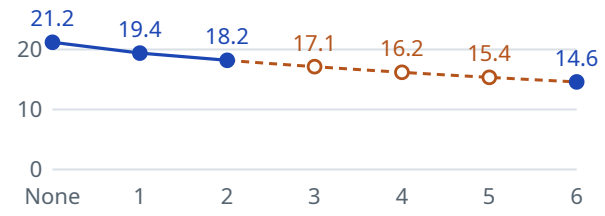
ours Per seahorse: body/tail 1,027 triangles + fins 74. Residents share meshes and atlas; males have a brood pouch. Yellow/orange sex colours are authored.

ours 7 three-angle knots + 4 wrap values drive 35 joints; clips blend and joint limits apply. The deformer block has 40 cells; behavior uses more.

reference check (ours) 7 exported-rig poses: no frame flips, ≤ 95.91% joint-limit use, ≤ 0.258 mm wrap-centre error. Python reference, not a device measurement.

08 · Six pairs: measured on cast1

Close idle · FPS



Main-thread CPU

6 pairs — None · close idle

+20.67 ms

+9.21 update
+11.47 render
animation +2.68
already inside update

measured (ours) Close idle: None → 6 pairs, 21.19 → 14.59 FPS. Crawl: 21.13 → 14.58 FPS. 3 × 60 s/count/view, 2026-10-17.

estimate (ours) 3–5 pairs: frame-time interpolation between measured 2 and 6.

other species *H. erectus* breeder guideline: one adult pair per 10 gal. [13]

ours Our 40 gal game defaults to 6 pairs. None retains 12 residents and hides 6 posts.

verified source read; it says this of *H. reidi* other species source read; it is about the species named ours game choices or measurements; estimates say **estimate (ours)**

[1] Hettiarachchi & Edirisinghe 2017. Trop. Agric. Res. 29: 77–89 · abstract · *H. reidi*

[2] Lourie, Foster, Cooper & Vincent 2004. A Guide to the Identification of Seahorses · pp. 2, 72–73 · *H. reidi*

[3] Silveira et al. 2014. Zootaxa 3861: 317–332 · pp. 322–323 · *H. reidi*

[4] Maia, Couto & Adriaens 2013. How seahorses hang on to their life. SICB abstract · abstract · *H. reidi, H. abdominalis*

[5] Porter et al. 2015. Why the seahorse tail is square. Science 349: aaa6683 · Figs. 1, 2, 5 · *H. reidi, H. kuda; printed models*

[6] Zima-Kulisiewicz et al. 2009. Undulatory fin motion of seahorse. GALA proceedings 56 · pp. 56-4–56-5 · *H. reidi*

[7] Botello-Payro et al. 2010. Flow induced by seahorse fins. GALA proceedings 28 · results · *H. reidi*

[8] Ashley-Ross 2002. J. Exp. Zool. 293: 561–577 · p. 566 · *H. hudsonius (= H. erectus), H. zosterae*

[9] Li & Xu 2016. Swimming ability of the yellow seahorse. Marine Sciences 40(5): 76–81 · abstract · *H. kuda*

[10] Roos et al. 2009. Suction feeding in H. reidi. J. Exp. Biol. 212: 3490–3498 · p. 3493, Table 1 · *H. reidi*

[11] Van Wassenbergh et al. 2009. Suction is kid's play. Biol. Lett. 5: 200–203 · p. 201, Fig. 2 · *H. reidi newborns*

[12] Mosk et al. 2007. Visual Neuroscience 24: 345–354 · abstract · *Syngnathidae*

[13] Ocean Rider (breeder). Stocking Density forum thread, 2006 · reply #2823 · *H. erectus*

Sources checked 2026-10-16; game measured 2026-10-17. Saltwater, seed 713, age 150, growth paused. Locators, DOIs and access limits: docs/reference/seahorse-research.md. The drawings are original and use the game's own ring table. Claims the research could not confirm stay in the research document. A3 portrait, 297 × 420 mm; print on matte or uncoated stock.